Dynasonics® Ultrasonic Flow Meters

IS-4000

Open-Channel Flow Meter

DESCRIPTION

An economical open-channel flow meter, IS-4000 measures level, flow rate and total volume of water flowing through weirs and flumes. The meter includes a non-contact ultrasonic level sensor to detect the water level and then calculates the flow rate and total volume based on characteristics of the channel. All the measurements are available over Modbus RTU or Modbus TCP Ethernet and can be logged for historical records.

BENEFITS

- Measure level, flow rate and total volume with a single device
- Simple setup for flumes and weirs
- Automatically retains an historical log of all measurements and configuration events
- Easily connect up to SCADA systems
- Connectivity with BEACON Advanced Metering Analytics (AMA) or AquaCUE Flow Measurement Manager

OPERATION

Based on empirical formulas, the IS-4000 calculates the flow rate based on the geometry of the channel or primary device and water depth. The level sensor measures the depth of the water used in the calculation.

The IS-4000 includes a selection of primary devices with preprogrammed tables to simplify the setup, including:

- Parshall flumes
- Manhole flumes
- V-notch weirs

Additionally, you can enter custom tables using the Flow Meter Tool software.

APPLICATIONS

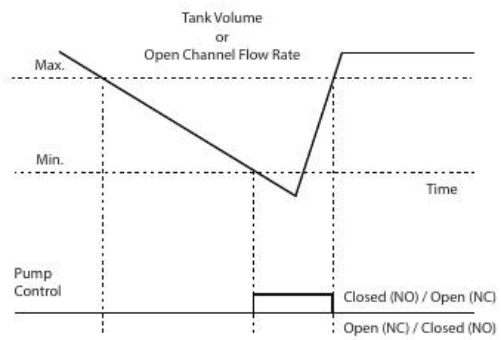
Open channels with a primary device are a cost effective solution for managing varying flow rates in unpressurized systems. The IS-4000 flow meter performs best when used with a primary device such as a flume or weir, and where the sediment does not build up. ③

- Flow into water treatment plants from reservoirs
- Storm and sanitary sewer systems
- Effluent from water resource recovery or wastewater treatment
- Industrial discharge
- Agriculture irrigation channels



PUMP CONTROL OPTION

The Pump Control option automatically starts and stops the pump based on water level.



SPECIFICATIONS

System Specifications

Liquid Types	Unpressurized liquids with minimal foam ① in open channels or partially filled pipes
Channel Selection	Weir: Contracted rectangular, suppressed rectangular, Cipoletti; V-notch weir (30°, 45°, 60°, 90°); ② Flume: Parshall (1, 2, 3, 6, 9, 12, 18, 24, 36, 48 and 60 in.); Manhole flume (4, 6, 8, 10 and 12 in.), Manning rectangle (up to 9.8 ft, 6 m) Pipes and Other: Manning pipe, exponential equation

(Specifications continued on next page)

Badger Meter

HYB-DS-02510-EN-09 (May 2025)

Product Data Sheet

1. น้ำต้องมีฟองน้อย
   => มีผนัง □ □ ตัดฟองแล้ว
2. รองรับ V-notch 90° มีสูตรในเครื่องแล้ว
3. ตะกอนต้องไม่สะสมหน้าเวียร์
   => ล้างรางเป็นระยะ

Flow meter — สเปกหัววัด DL10 (Data sheet หน้า 2)

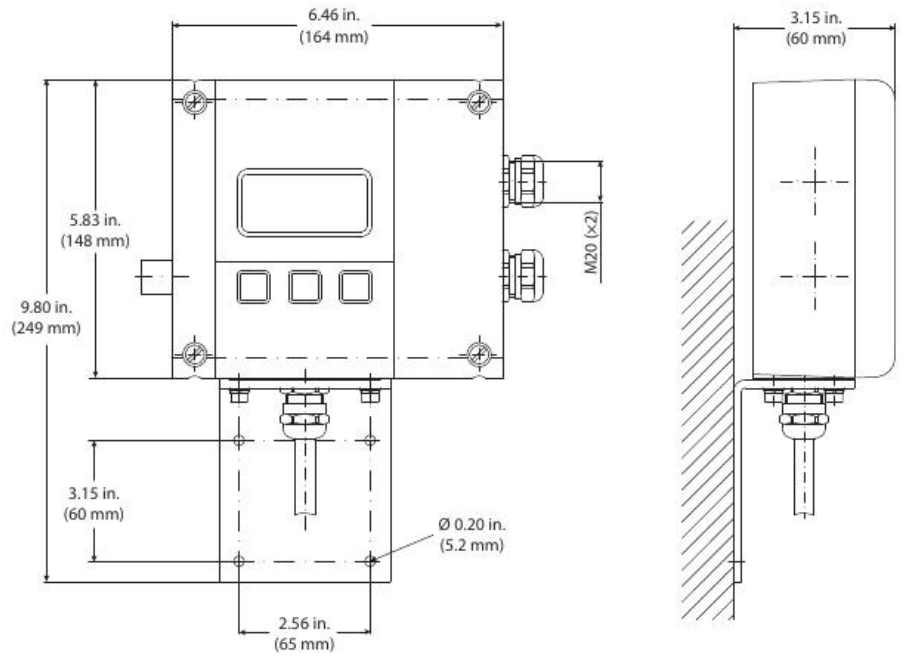
IS-4000, Open-Channel Flow Meter

Electronics Specifications

Power	92…275V AC (50/60 Hz), < 14 VA
Display	Graphical LCD 64 × 128, backlight, actual flow rate, totalizers, status display
Configuration	3 front-panel mounted push-buttons or mini USB with IP67 connector included
Enclosure	Die cast powder-coated aluminium, protection class IP67
Cable Connection	Supply and signal cables 2 × M20; cable glands included From meter M20; cable gland included
Environmental	–4…140° F (–20 up to 60° C)
Analog output	4…20 mA, 0…20 mA, 0…10 mA ≤ 800 Ohm, active or passive; Assigned parameter depends on flow meter mode
Level sensor input	4…20 mA from level sensor
Digital outputs	2 open collectors; passive: maximum 32V DC, 0…100 Hz 100 mA, 100…10.000 Hz 20 mA; active: 24V DC, maximum 20 mA; Select active pulse (up to 2000 msec), minimum/maximum alarm, error messages or pump control
	Solid-state relay (n.o./n.c.) maximum 230V AC, 500 mA, 1 Hz; Function is linked with open collector output 2
Digital input	5…30V DC; totalizer reset, positive return zero, BEACON/AquaCUE connectivity
Communication	RS485 Modbus RTU, Modbus TCP/IP Ethernet, BEACON/AquaCUE connectivity
Programming port	Mini B USB, IP67
Datalogger	2 MB capacity with 130,000 logged lines: date, level, flow rate, tank volume
Security	Three-level password
Languages	English, French, German, Italian, Spanish, Czech, Russian
Certification	CE Low Voltage Directive 2014/35/EU, EMC 2014/30/EU, RoHS 2006 2011/65/EU, 2015/863/EU, 2017/2102/EU

Sensors Specifications

Sensor Type	DL 10 Ultrasonic	DL 24 Ultrasonic	ULM 53 Ultrasonic	ULM 70 Ultrasonic	C 21 Radar
Measuring Range	0…49.21 in. (0…1250 mm)	0…9.8 ft (0…3000 mm)	0…19.7 ft (0…6000 mm)	0…6.6 ft (0…2000 mm)	0…49.2 ft (0…15000 mm)
Beam Width	2°	2°	14°	10°	8°
Accuracy	0.125 in. (3 mm)	0.25 in. (6 mm)	0.35 in. (9 mm)	0.125 in. (3 mm)	0.08 in. (2 mm)
Deadband	2 in. (50 mm)	4 in. (100 mm)	8 in. (200 mm)	6 in. (150 mm)	9.84 in. (250 mm)
Ambient Temperature	–31…140° F (–35…60° C)	–31…140° F (–35…60° C)	–22…158° F (–3…70° C)	–22…158° F (–3…70° C)	-40…176° (-40…80°)
Transducer Material	PVDF	PVDF	PVC/PVDF	PVC/PVDF	PVDF
Protection Class	Type 6P submerged 6 feet (1.8 m) of water, up to 24 hr	Type 6P submerged 6 feet (1.8 m) of water, up to 24 hr	IP68	IP67	IP66/IP68, Type 4X/6P
Mount (US)	1 in. NPT	1 in. NPT	—	—	1-1/2 in. NPT
Mount (EU)	G1	G1	G 1-1/2	G 1-1/2	G 1-1/2
Ratings	CE, RoHS	CE, RoHS	CE (LVD, EMC, RoHS)	ATEX II 2G Ex ia IIB T5 Ga/Gb with isolator	CE (EMC, LVD, RED, RoHS), UKCA
Dimensions H × W × D	3.2 × 2.0 × 2.0 in. (81 × 51 × 51 mm)	4.9 × 3.1 × 3.1 in. (122 × 78 × 78)	5.1 × 2.2 × 2.2 in. (129 × 55 × 55 mm)	4.8 × 2.8 × 28 in. (121 × 71 × 71 mm)	4.28 × 2.99 × 2.99 in. (109 × 76 × 76 mm)



1. รุ่นที่ใช้ = DL10 (คอลัมน์นี้)
2. วัดได้ไกลสุด 1.25 m
   => หัววัดห่างยอด V ไม่เกิน 1.25 m
3. คลื่นแคบ => ติดกลางราง ใต้หัววัดโล่ง
4. แม่นยำ +/-3 mm
5. ระยะบอด 5 cm => หน้าหัววัดต้องสูงกว่า
   น้ำสูงสุด >= 5 cm (แนะนำ 20–30 cm)
6. เกลียวยึด G1 (1 นิ้ว)

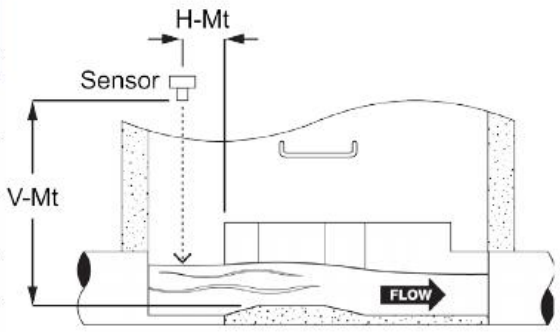
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Mounting Positions

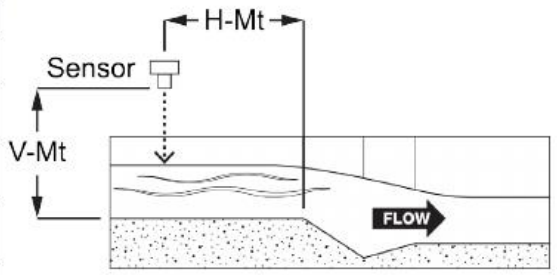
Manhole Flume

Size	Max. Flow	Max. Water Level	V-Mt	H-Mt
in. (DN)	g/sec (l/sec)	in. (mm)	in. (mm)	in. (mm)
4 (100)	1.32 (5)	5.83 (148)	23.62 (600)	5.75 (146)
6 (150)	4.23 (16)	8.94 (227)	23.62 (600)	7.75 (197)
8 (200)	9.25 (35)	12.28 (312)	23.62 (600)	9.76 (248)
10 (250)	16.64 (63)	15.55 (395)	27.56 (700)	11.73 (298)
12 (300)	24.83 (94)	18.00 (457)	27.56 (700)	13.74 (349)

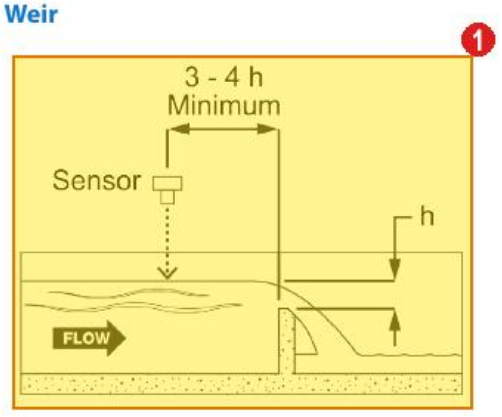


Parshall Flume

Size	Max. Flow	V-Mt	H-Mt
in. (mm)	g/sec (l/sec)	in. (mm)	in. (mm)
3 (75)	14.26 (54)	30.71 (780)	12.00 (305)
6 (150)	30.12 (114)	30.71 (780)	15.98 (406)
9 (230)	77.67 (284)	38.19 (970)	22.52 (572)
12 (305)	157.98 (598)	contact factory	contact factory
18 (455)	24.83 (94)	contact factory	contact factory



Weir



1 หัววัดอยู่ต้นน้ำ ห่างแผ่นเวียร์
>= 3–4 h (h = ความสูงน้ำเหนือยอด V)
h สูงสุด 0.21 => >= 0.63–0.84 m
=> ใช้ประมาณ 0.9 m

Meter Setup Menu

Application	**Tank** 	Select for a tank application.
	Open Channel 	Select for an open channel application.
Sensor	**Interval** 	Setup of time measurement interval(s); default value is 1 second; larger interval (for instance, 300 seconds) is set when unit is powered from battery.
	WarmUpTime 	Powering time of sensor(s) before measurement; larger interval is set when unit is powered from battery
	LowerRangeValue 	The minimum level value of used sensor = 4 mA in selected level units For most level sensors, set to zero.
	UpperRangeValue 	The maximum level value of used sensor = 20 mA in selected level units For a DL10 sensor, set to 49.2 in. (1250 mm) For a DL24 sensor, set to 118.1 in. (3000 mm) For a ULM 53, set to 9.8 ft (3 m) For a ULM 70, set to 6.6 ft (2 m) For a C 21, set to 49.2 ft (1500 mm) Set the units in the Length parameter in the Measurement Menu.
	Offset 	Level offset in selected level units, depends on sensor mounting position If the sensor is mounted lower than the specified height, enter the difference as a negative offset. To calculate the offset, measure the distance from the zero flow position and subtract the maximum height specification of the level sensor. For example on a flume or partial pipe, if a DL10 level sensor (max height 1250mm) is mounted 600 mm above the bottom of the channel, the offset is -650 mm. After setting the offset, check the level reading on the first screen with no flow in the channel. If necessary, adjust the offset value in small increments until the level measurement is 0. For a weir, the zero flow position is the bottom of the notch, not the bottom of channel. It is normal for the level measurement to be negative when the water level is below the notch or when the channel is dry.

1. Offset = ระยะหัววัดถึงจุดศูนย์ □ 1,250 mm (ค่าติดลบ)
2. เวียร์: จุดศูนย์ = ยอด V ไม่ใช่พื้นราง
3. น้ำต่ำกว่ายอด V จออ่านติดลบ = ปกติ

TROUBLESHOOTING

The following error messages may display:

Description	Possible Cause	Recommended Action
Pulse Output	Pulse rate exceeds the maximum	Reduce pulse scale (pulse/unit) and/or reduce pulse width configuration
EEPROM	Configuration file is missing	Replace transmitter
Configuration	Configuration file is corrupted	Update firmware. Replace transmitter if error repeats
Low Battery	Low backup battery (memory)	Replace transmitter
Measure Timeout	Measurement was not completed within specific time	Increase the WarmUpTime in Meter Setup Menu > Sensor menu. Check level sensor operation and wiring

The following symptoms may occur:

Symptom	Possible Cause	Recommended Action
Flow measurement is indicated when there is no actual flow	Offset is not programmed correctly. Level sensor mounting needs adjustment.	Verify the level reading is 0. If using a Weir, the level reading may be negative when the Weir is dry or below the notch of edge. Verify the Sensor > Offset is entered as a negative number. Verify the level sensor is aimed at the correct location in the channel. Consult the channel specifications for the correct placement of the level sensor. Verify the level sensor is not too close to walls or other structures that could interfere with the signal. Check the Beam Width and Deadband of the level sensor specification. If Sensor > UpperRangeValue is correct for the level sensor, adjust the Offset until the level reads 0.
Flow reading fluctuates	Level sensor mounting needs adjustment. Debris or condensate on level sensor. Electrical noise interference.	Verify the level sensor and check for interference with beam path, including floating debris, support bars and spider webs. Verify the level sensor beam reflects off a steady smooth surface without waves. Install a stilling well if wind or other disturbances are frequent. Verify the level sensor is aimed at the correct location in the channel. Consult the channel specifications for the correct placement of the level sensor. Verify the level sensor is not too close to walls or other structures that could interfere with the signal. Check the Beam Width and Deadband of the level sensor specification. Check the ground connections in the IS-4000. Ground shield of the cable between the IS-4000 and level sensor.
IS-4000 does not power up	No power or inadequate power.	Measure voltage at the power terminals and verify the voltage matches the labels by the power terminals. Verify the control LED 10 is on. If light is on and display does not power up, replace IS-4000.
Current or pulse outputs do not match readings	Parameter settings do not match control system. Wiring is incorrect.	The analog output is scaled from 0 flow at 0 mA or 4 mA to the distance between the 0 flow level and the bottom of the level sensor (maximum level sensor height less the offset). Program the control system the same. For pulse outputs, check the settings in Inputs/Outputs > Digital Outputs and compare to the control system.

1. อย่าติดชิดผนังหรือโครงสร้าง
2. ผิวน้ำใต้หัววัดต้องนิ่ง เรียบ
3. มีลม/คลื่น => ทำท่อพักน้ำนิ่ง (stilling well)

DATASHEET D2.16

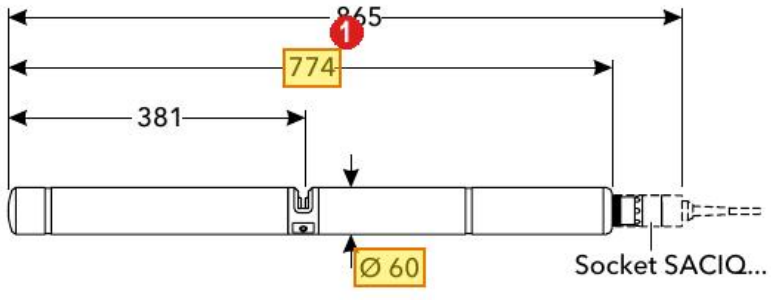


Optical SAC and UVT sensor UV 70x IQ SAC

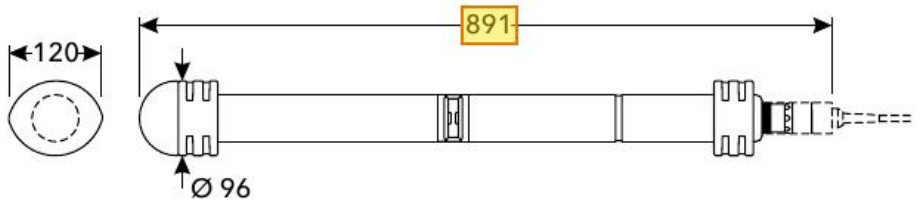
Low-cost probe (integrated ultrasonic cleaning, turbidity compensation) for the maintenance-free and reagent-free SAC measurement according to DIN 38404 C3

We would like to inform you about the application range on our website

UV 701 IQ SAC, UV 705 IQ SAC



With shock protection:



1. ยาว 774 mm (891 รวมปลอกกันกระแทก)
   Ø60 mm
2. ปรับมาสำหรับน้ำเสียชุมชน
   => ต้องสอบเทียบกับผล BOD แล็บ
3. รุ่น 705 วัด BOD ได้ 0–500 mg/L
4. ความเร็วน้ำ <= 3 m/s (ช่อง 3 ~ 0.02 (ผ่าน))
5. แรงดัน <= 1 bar (จมลึกได้ ~10 m)
6. หนัก 4 kg => ขาจับต้องแข็งแรง
7. ไม่รวมสาย! ต้องสั่งสาย SACIQ แยก
   + ชุดควบคุม IQ Sensor Net

Technical Data

Model		UV 701 IQ SAC	UV 705 IQ SAC
Measuring method		UV-Absorptionsmessung 254 nm (Kompensation 550 nm)	
Measuring gap (optical layer thickness)		1 mm	5 mm
Application (optimized for)		Municipal wastewater with a low proportion of industrial wastewater,	wastewater treatment plants, surface water
Measuring range and Resolution	COD	0.0 ... 12,500 mg/l 1 mg/l	0.0 ... 800 mg/l 1 mg/l
	TOC	0.0 ... 20,000 mg/l 1 mg/l	0.0 ... 500.0 mg/l 0.1 mg/l
	DOC	0.0 ... 12,500 mg/l 1 mg/l	0.0 ... 500.0 mg/l 0.1 mg/l
	BOD	0.0 ... 8,000 mg/l 1 mg/l	0.0 ... 500.0 mg/l 0.1 mg/l
	$SAC_{254\ total}$	0.0 ... 3,000 1/m 1 1/m	0.0 ... 600.0 1/m 0.1 1/m
	$SAC_{254\ dissolv}$	0.0 ... 3,000 1/m 1 1/m	0.0 ... 600.0 1/m 0.1 1/m
	$UVT_{254\ total}$*	0.0 ... 100.0 % 0.1 %	0.0 ... 100.0 % 0.1 %
	$UVT_{254\ dissolv}$*	0.0 ... 100.0 % 0.1 %	0.0 ... 100.0 % 0.1 %
Flow rate		≤ 3 m/s	
Pressure Resistance		Maximum 1 bar (incl. sensor connection cable)	
Electrical connections		2-wire shield cable with quick fastener to sensor	
Electromagnetic Compatibility		EN 61326, Class B, FCC Class A Intended for indispensable operation	
Certifications		CE	
Mechanical		Housing: Titan Grade 2, PEEK Window: Sapphire glass Protection class: IP 68	
Weight (without cable)		Approx. 8.82 lb (4 kg)	
Warranty		2 years for defects in quality	

** The UVT-254 value is standardized to 10 mm gap width.*

Model	Description	Order No.
UV 701 IQ SAC	Optical SAC and UVT sensor (254 nm) to measure higher concentrations with integrated ultrasonic cleaning, multifunctional slide and shock-absorption-rings, without connecting cable (order SACIQ separately)	481036
UV 705 IQ SAC	Like UV 701 IQ SAC, but to measure lower concentrations	481038

WTW a xylem brand

Xylem Analytics Germany Sales GmbH & Co. KG, WTW **www.WTW.com**
Dr.-Karl-Slevogt-Straße 1 · D-82362 Weilheim · Germany · Phone: +49 881 1830 · Fax: +49 881 183-420 · Info.WTW@Xyleminc.com

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